

Pythagoras Theorem - Problems

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Solve these problems using Pythagoras Theorem, answer to 2 decimal places.



- 1 A 2.3 m ladder is resting against a wall as shown. If the foot of the ladder is 0.8 m from the base of the wall, how far up the wall will the ladder reach? Start by filling in the boxes.

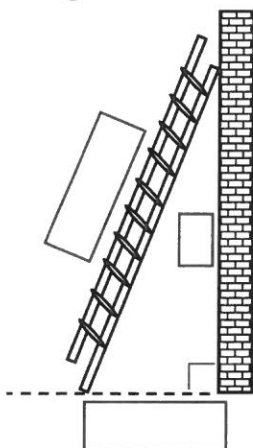
$$h^2 =$$

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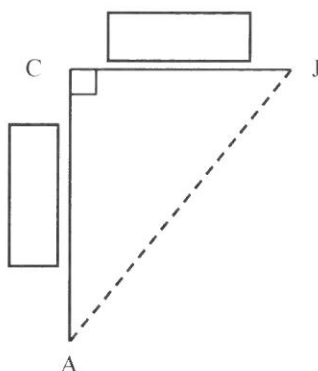
$$h = \sqrt{\quad}$$

$$h =$$

The ladder will reach



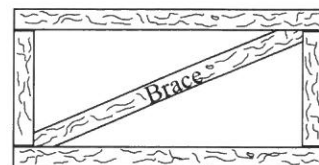
- 2 Jake and Amy left the same campsite (C). Jake (J) walked 4.2 km East while Amy (A) walked 5.6 km South. How far apart are they?



Answer using a sentence

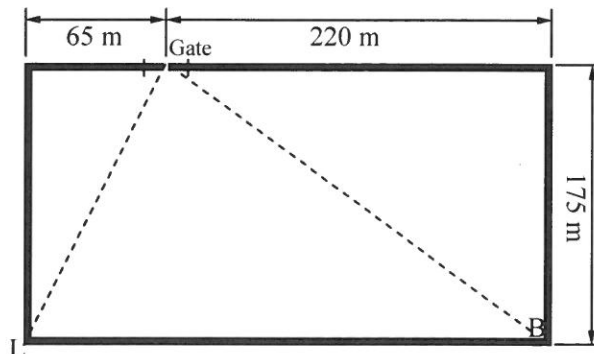
- 3 A kite is on a 35 m string and is a horizontal distance of 22 m from the flier. How high is the kite above the ground? Construct a diagram showing all relevant details.

- 4 A driveway gate (3.4 m × 1.2 m) is shown, calculate the length of the wooden brace.



- 5 Calculate the cost to build the gate given that the wood costs \$7.30 per metre.

- 6 Leaping Larry always takes a shortcut through Farmer Fred's paddock. Bezerko the bull chases Larry because Larry wears red socks. If Larry is at point L and Bezerko is at B, how far does each have to run to get to the gate?



Larry:

Bezerko:

- 7 Find Larry's distance as a percentage of Bezerko's, round to the nearest whole number.
